

# DIN 5480 Spline Insertion Dynamics

## VBD vs MuJoCo Solver Benchmark

*Core question:*

Which solver correctly captures tooth-groove engagement?

What physical mechanisms explain the divergence?

**6-tooth**

DIN 5480  $z=6$

**23 s**

scripted trajectory

**2 solvers**

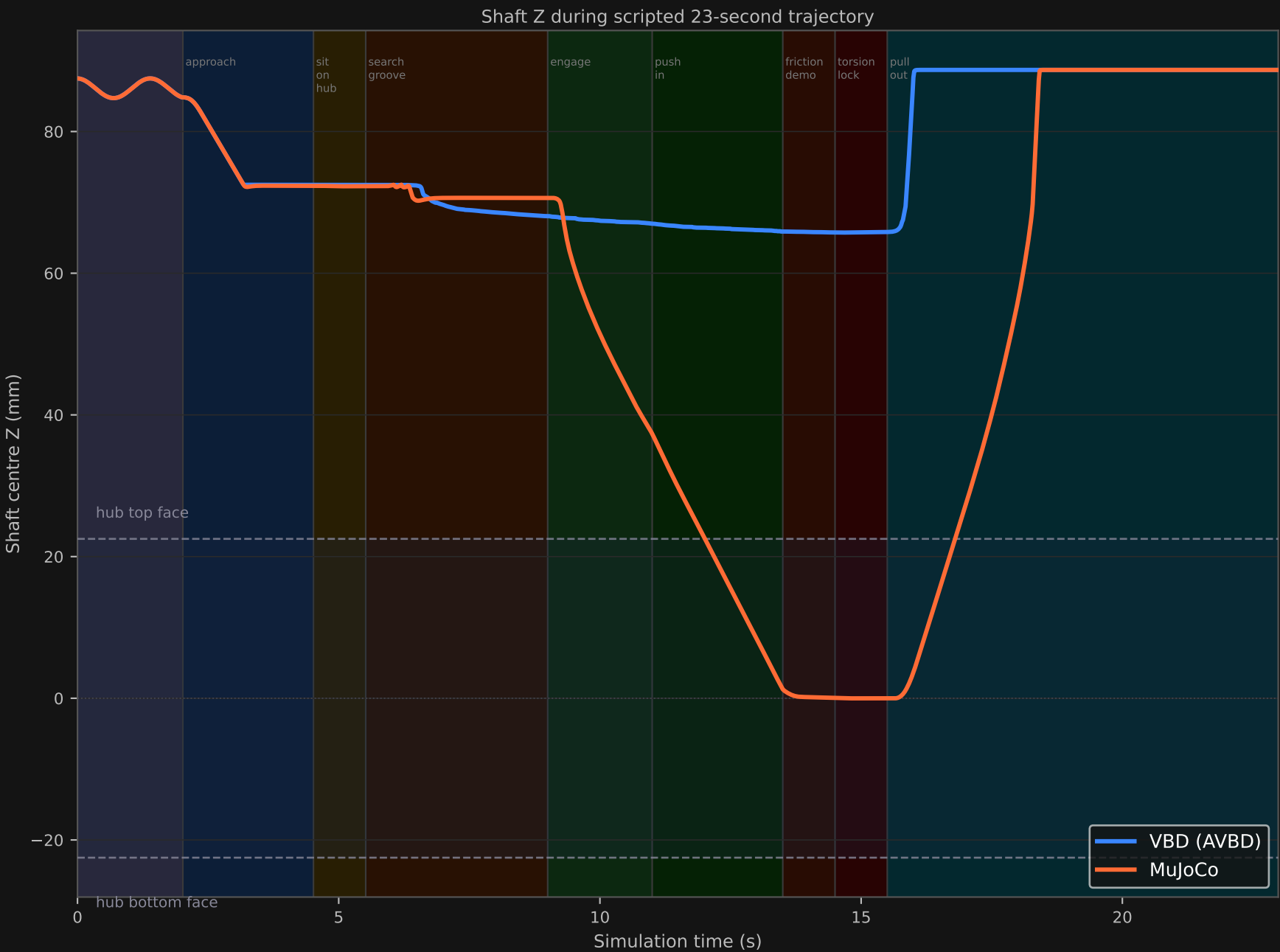
VBD / MuJoCo

**Auto-benchmark**

headless, repeatable

# The Task: Simulate Search-and-Engage Insertion

DIN 5480 · involute spline · z=6, module=4 mm, 30° pressure angle · hub bore  $\pm 45$  mm



## What we simulate

- ▶ Shaft descends into hub bore under P-controller descent force
- ▶ Teeth search for groove alignment while rotating slowly
- ▶ Hydroelastic contact on all tooth and land faces simultaneously
- ▶ Spring-damper on-axis guidance (lateral + rotation + descent)

## Why it's physically hard

- ▶ Teeth ride on land faces → high resistance, near-zero insertion
- ▶ Groove alignment triggers a snap-to-engage drop in force
- ▶ Solver must resolve this snap in real time

## What we measure

- ▶ Shaft position, contact area, penetration depth, frame time
- ▶ Engagement time, snap detection, torsion-lock stiffness

# Auto-Benchmark: Fair Comparison Design

Parameters identical except where the contact algorithm forces a difference · headless, repeatable, same trajectory for both

## ✓ KEPT IDENTICAL — same inputs, same conditions

|                             |                                              |
|-----------------------------|----------------------------------------------|
| ✓ <b>Geometry:</b>          | DIN 5480 z=6, module=4 mm, 40 mm length      |
| ✓ <b>Trajectory script:</b> | Same scripted_trajectory.py (23 s, 9 phases) |
| ✓ <b>Shaft mass:</b>        | 1.00 kg · solid cylinder inertia             |
| ✓ <b>Contact pipeline:</b>  | Newton hydroelastic SDF (both solvers)       |
| ✓ <b>Descent gain:</b>      | descent_k = 800 N·s/m                        |
| ✓ <b>Rotation damping:</b>  | rot_d = 0.10–0.13 N·m·s/rad                  |
| ✓ <b>Lateral spring:</b>    | lateral_k = 1×10 <sup>5</sup> N/m            |
| ✓ <b>Friction coeff.:</b>   | μ = 0.30 (dry steel-on-steel)                |
| ✓ <b>SDF resolution:</b>    | 64 <sup>3</sup> voxels, narrow-band          |
| ✓ <b>Warm-up frames:</b>    | 5 frames excluded from all timing stats      |

## △ MUST DIFFER — contact algorithm forces the difference

### △ Substeps / frame:

VBD = 8      MuJoCo = 16

*MuJoCo force-level model needs tighter sub\_dt (1 ms)  
for per-substep contact correction during fast rotation*

### △ Contact model:

VBD = position-level (AVBD)  
MuJoCo = force-level (kh × volume)

*AVBD directly corrects penetration each sub-step;  
force model needs stiffness to generate resisting force*

### △ Contact stiffness kh:

VBD = 5×10<sup>6</sup>    MuJoCo = 2×10<sup>9</sup>

*Force-level equilibrium: δ\_eq = F / (kh · A)  
400× stiffer MJ kh gives <2 mm equilibrium penetration*

### △ Collision frequency:

VBD = once per frame  
MuJoCo = every substep

*Force model needs fresh geometry at every sub-step  
to correctly block a rotating + descending shaft*

# Results: Insertion Trajectory & Performance

Same scripted trajectory, same inputs — solver contact model is the only difference



## VBD (AVBD)

6.8 mm

max insertion depth

84 ms/fr · 0.20× RTF

## MuJoCo

72.5 mm

max insertion depth

114 ms/fr · 0.15× RTF

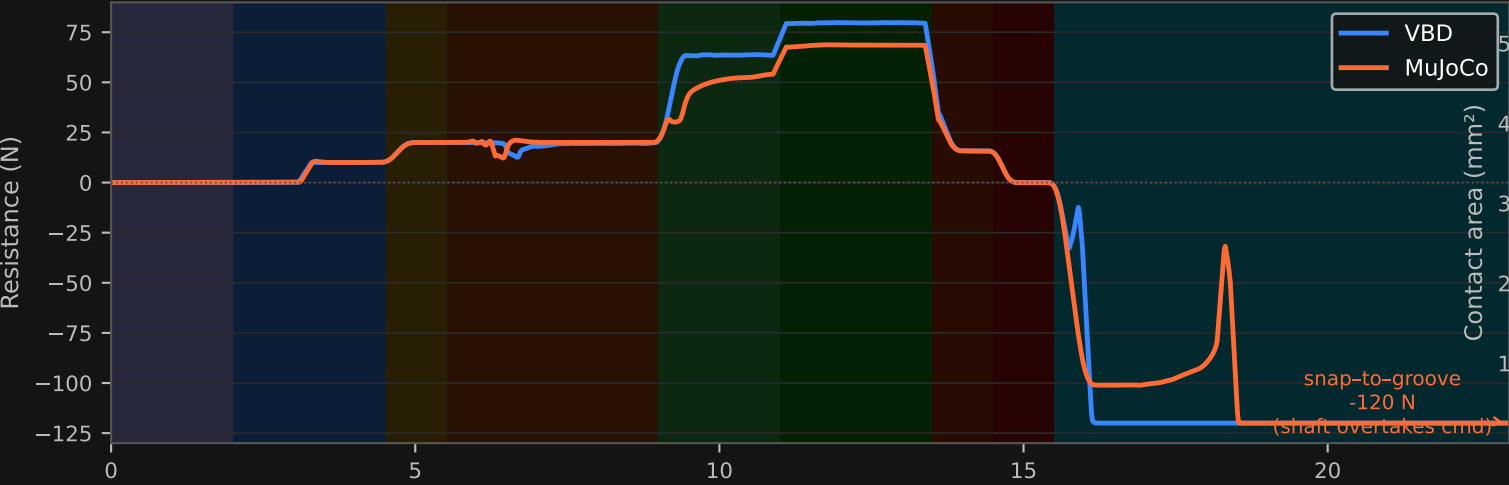
## Key insight

VBD per-substep is 1.47× faster. MuJoCo inserts correctly because force-level contacts let the solver converge to an equilibrium state rather than blocking penetration.

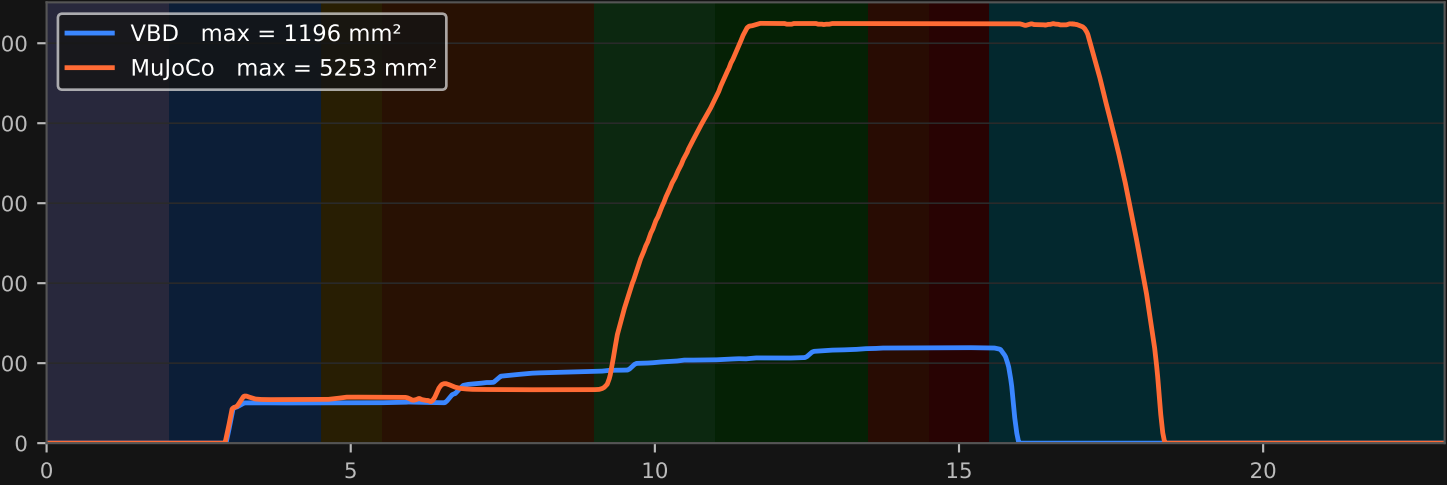
# Contact Forces, Area & Convergence

Hydroelastic pipeline metrics · resistance proxy = descent\_k × velocity error · kh VBD=5×10<sup>6</sup> MuJoCo=2×10<sup>9</sup>

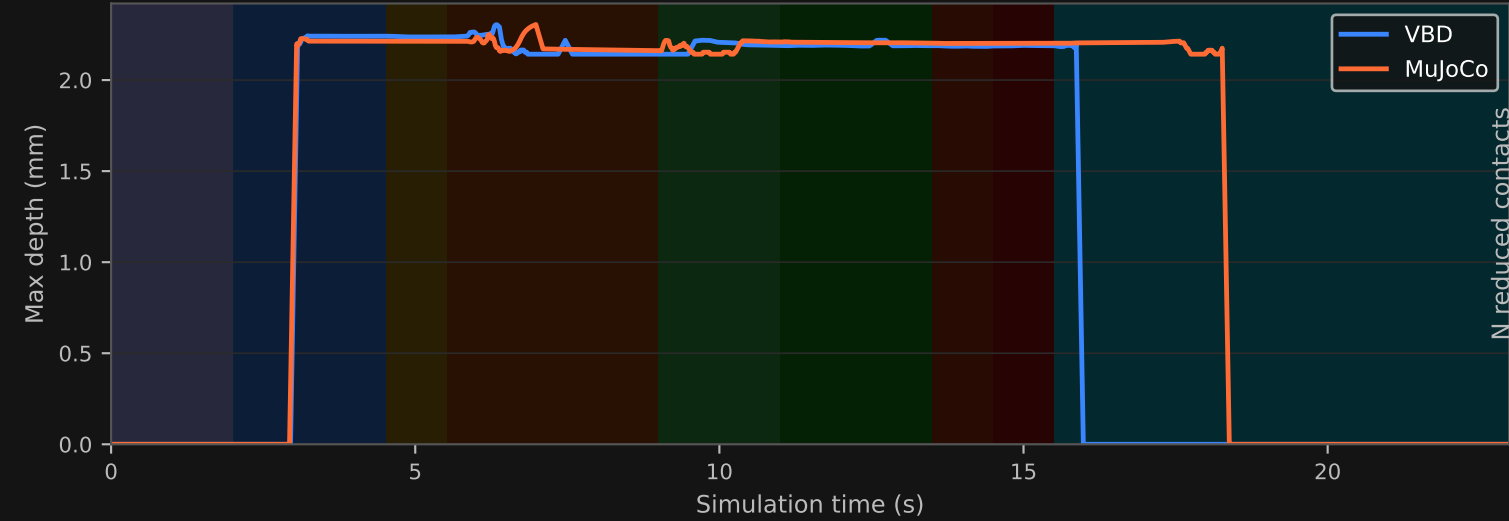
Axial contact resistance = descent\_k × (actual vel – commanded vel) · +ve = blocking descent



Hydroelastic contact patch area



Max penetration depth · AVBD correction proxy: near-0 = fully corrected



Reduced contact point count



VBD/AVBD: position-correction drives  $\delta \rightarrow 0 \rightarrow$  stiff tooth faces block descent  $\rightarrow$  small contact area, near-zero depth, high resistance while teeth ride lands  
MuJoCo: force-level equilibrium ( $\delta_{eq} = F/kh/A$ )  $\rightarrow$  shaft descends through groove  $\rightarrow$  contact area grows 4×  $\rightarrow$  snap event visible in resistance (overtakes command)